

BCI-Based Robotic Arm for Medical Rehabilitation

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Abstract—Stroke and other neurological disorders often result in significant motor impairments, limiting the effectiveness of traditional rehabilitation. This project proposes a non-invasive Brain-Computer Interface (BCI) system designed to assist motor rehabilitation by decoding motor imagery (MI) tasks from EEG signals. Using the publicly available PhysioNet EEG Motor Imagery dataset, EEG signals from only three strategically selected electrodes (C3, Cz, C4) over the sensorimotor cortex are utilized, significantly reducing hardware complexity without compromising performance. Signals from 30 subjects are pre-processed to remove noise and artifacts, then transformed into 2D spectrograms via Short-Time Fourier Transform (STFT) to capture time-frequency features. A hybrid deep learning model is employed, where a Convolutional Neural Network (CNN) extracts spatial features from the spectrogram, and a Long Short-Term Memory (LSTM) network models the temporal dynamics for accurate classification of imagined movements (left, right and rest). The classified outputs are then translated into control commands and transmitted to an Arduino Uno microcontroller, which actuates a 4-DOF robotic arm using SG90 servo motors. This closed-loop system demonstrates a novel, low-cost, and adaptive BCI framework for hands-free robotic assistance and neurorehabilitation therapy. By leveraging an optimal number of subjects and selecting only three key motor cortex electrodes (C3, Cz, and C4), the system achieves effective motor intention decoding with reduced computational and hardware complexity. The deep learning model achieves a test accuracy of 85.89%, making it suitable for scalable and practical BCI deployment.

Index Terms—Brain Computer Interface, CNN-LSTM, Rehabilitation, Electroencephalography (EEG)

I. INTRODUCTION

Motor impairments resulting from neurological conditions such as stroke, spinal cord injury, or neurodegenerative diseases often lead to partial or complete loss of motor function, significantly affecting a patient's independence and quality of life. Traditional rehabilitation methods primarily rely on voluntary physical movement, which may not be feasible for individuals with severe paralysis. In this context, Brain-Computer Interface (BCI) systems offer a promising alternative by establishing a direct communication pathway between the brain and external devices, enabling control without physical movement.

In this work, we present a real-time BCI-based robotic arm control system tailored for motor rehabilitation. The system leverages non-invasive electroencephalography (EEG) signals to interpret motor imagery (MI) tasks, such as imagining left or right hand movement, and translates them into control commands for a robotic arm. Using the publicly available PhysioNet Motor Imagery Dataset [1], which contains EEG recordings from multiple subjects, we process and classify neural activity related to imagined movement.

Several recent studies have advanced BCI-based systems for motor imagery classification and rehabilitation. Fadel et al [2] proposed a deep recurrent convolutional neural network that transformed EEG signals into 2D image representations using azimuthal equidistant projection and frequency band interpolation, achieving strong multi-class motor imagery classification accuracy. Li et al. [3] proposed a hybrid neural network integrating CNN and BiLSTM to improve EEG-based motor imagery classification through spatial-temporal feature fusion. Satam and Szabolcsi [4] developed a 6-DOF robotic arm controlled via a neurofeedback-driven Arduino-based BCI system, enabling real-time motor control. Yamamoto and Hamaguchi [5] demonstrated the feasibility of a BCI-enabled upper-limb assistive device for active rehabilitation. Alnaanah et al. [6] evaluated various CNN architectures on the PhysioNet dataset, showing that frequency-specific filtering improves cross-subject generalization. These studies collectively highlight the growing potential of deep learning and real-time EEG decoding in practical, cost-effective neurorehabilitation systems.

The paper is structured as follows: Section 2 outlines the system architecture and key components. Section 3 details the dataset and preprocessing steps, including filtering and spectrogram generation. Section 4 describes the CNN-LSTM model and training procedure. Section 5 covers the hardware setup, including the robotic arm and integration of the microcontroller. Section 6 presents performance metrics and results. Section 7 discusses the ablation study on CNN-LSTM model. Section 8 discusses findings in comparison with related

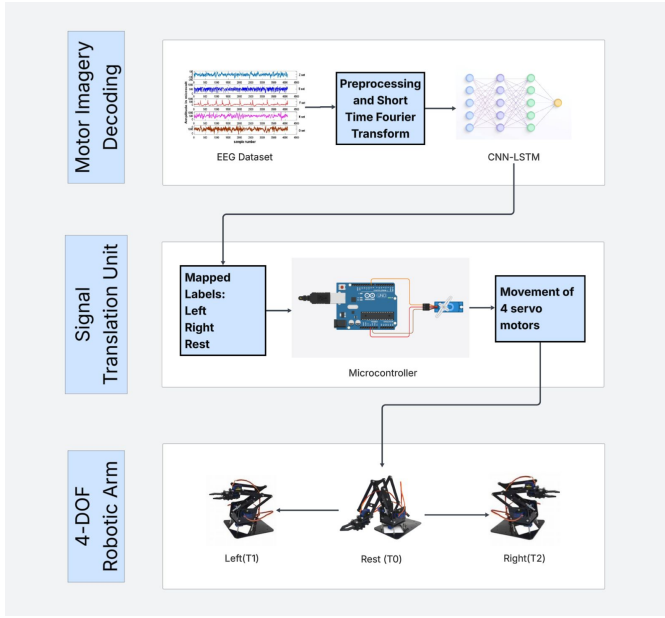


Fig. 1. System Overview

works. Section 9 concludes the study and suggests future directions.

II. SYSTEM OVERVIEW

Fig. 1 represents the overall architecture of the proposed system. The system comprises three primary components: (1) a deep learning-based motor imagery classification model, (2) a 4-DOF robotic arm for physical actuation, and (3) the integration interface that connects the trained model to the hardware via serial communication.

A. Deep Learning Motor Imagery Classification Model

The EEG signals used for classification are sourced from the PhysioNet Motor Imagery Dataset first 30 valid subjects data is used to train the model. These signals are preprocessed using band-stop and bandpass filtering to remove artifacts such as powerline noise and non-relevant frequency components. Each trial is segmented and converted into time-frequency spectrograms using the Short-Time Fourier Transform (STFT), with outputs from three key electrodes (C3, Cz, C4) stacked as RGB channels. These spectrograms are then fed into a CNN-LSTM hybrid architecture. The CNN acts as a spatial feature extractor, producing a 128-dimensional representation, which is sequentially passed to an LSTM network to capture temporal dynamics and classify the motor imagery task into categories such as left, right, or rest.

B. Robotic Arm

The hardware component consists of a 4-degree-of-freedom robotic arm actuated by SG90 servo motors. The arm mimics upper-limb movements such as lifting, rotating, grasping, and releasing. Each joint is mapped to a specific classification output, with motion constraints set to prevent overextension. The robotic arm is controlled by an Arduino Uno microcontroller,

which translates incoming movement commands into PWM signals for servo actuation.

C. Integration

The classified output from the LSTM model is transmitted via serial communication from the MATLAB environment to the Arduino. Each label (e.g., "Left", "Right", "Grasp", "Rest") is mapped to a corresponding movement routine on the Arduino, triggering synchronized servo motions in the robotic arm. This real-time closed-loop system enables the user to control the robotic arm solely through motor imagery, offering potential applications in neurorehabilitation by facilitating movement through cognitive intent.

III. DATASET PREPROCESSING

Pfurtscheller et al [7] emphasize the importance of C3, Cz, and C4 electrodes located over the sensorimotor cortex, as they effectively capture Event-related desynchronization/synchronization (ERD/ERS) [8] during motor tasks and imagery. Although the PhysioNet dataset includes 64 electrodes, C3, Cz, and C4 are chosen for their direct relevance to motor cortex activity. This focused selection simplifies processing while retaining key neural signals for accurate motor imagery classification.

EEG data are highly sensitive and prone to various artifacts such as movement disturbances, power line interference, and physiological noise. To ensure accurate feature extraction and effective model training, a series of pre-processing steps are applied to clean and standardize EEG signals. These steps include notch filtering, band pass filtering, and normalization to generate spectrograms for model input.

A. Notch Filtering

A Notch filter is applied at 60 Hz to eliminate power line noise commonly introduced by electrical grids in certain regions (e.g., North America). This frequency is chosen after observing the spectral distribution of the raw EEG signals plotted in Fig. 2. Filtering out this narrow band ensures the suppression of consistent, unwanted electrical interference without distorting the neural information.

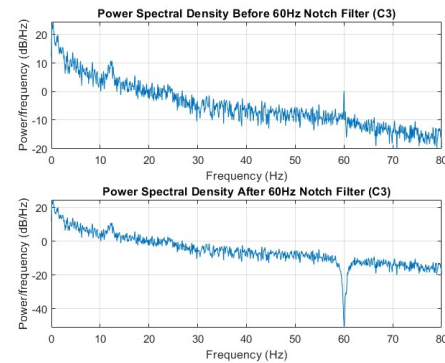


Fig. 2. Representation of power spectral density before and after for C3

B. Bandpass Filtering

To isolate the frequency range associated with motor imagery tasks, a bandpass filter is used. This filter allows only frequencies between 8 and 30 Hz to pass through, corresponding to the mu and beta rhythms of the brain, which are key to motor-related neural activity. Frequencies outside this range are attenuated to reduce the impact of irrelevant or noisy components. In this implementation, a 4th order Butterworth bandpass filter is applied between 8 and 30 Hz using an IIR design with a quality factor of 35 to suppress powerline interference. The difference before and after bandpass filtering is plotted in Fig. 3.

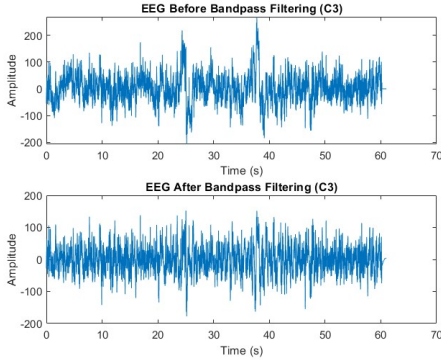


Fig. 3. Bandpass Filtering

C. Normalization

Fig. 4 illustrates normalization, which is applied to standardize the EEG signal across different channels and trials. This step transforms the data to zero mean and unit variance, making it more stable for numerical computations. It ensures that the model focuses on the patterns and features of the signal rather than the magnitude, leading to improved convergence during training, especially in deep neural networks.

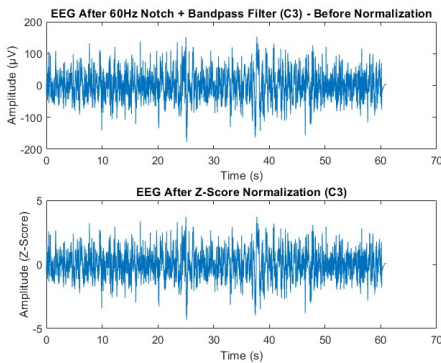


Fig. 4. Power spectral density before and after notch filtering for representative electrode C3

SPECTROGRAM CONVERSION

EEG signals from three electrodes (C3, C4, Cz) are pre-processed and segmented into fixed-length windows. Each

channel's time-series signal is transformed into a time-frequency representation using the Short-Time Fourier Transform (STFT), producing a spectrogram. In this process, a Hamming window of length 64 samples with 32-sample overlap and an FFT size of 128 is applied to extract localized frequency content. The magnitude of the spectrogram is computed using the absolute value and log-scaled to compress dynamic ranges. These individual spectrograms for the three channels are resized to a fixed dimension (224×224) and stacked along the third axis to form a three-channel (RGB-like) spectrogram image, as suggested in [9]. Fig. 5 illustrates this conversion pipeline. This stacked image effectively encodes spatial (channel-wise) and temporal frequency information, making it suitable for convolution neural networks to extract spatial patterns relevant to motor imagery classification.

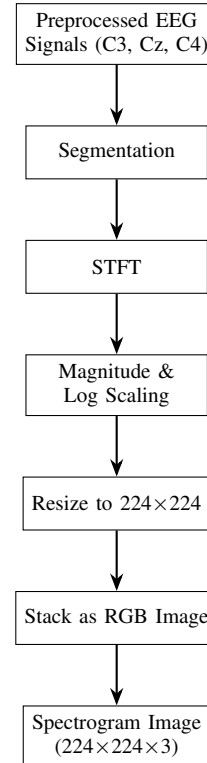


Fig. 5. Block diagram of EEG to Spectrogram Conversion Pipeline.

IV. MODEL ARCHITECTURE

A deep learning architecture that integrates CNN and LSTM networks to decode motor imagery EEG signals from the dataset and convert them into command for actuation of robotic arm.

A. CNN Architecture

The CNN in the proposed model is designed as a spatial feature extractor from the spectrogram images derived from EEG signals. The network starts with an input layer configured for $224 \times 224 \times 3$ RGB images. It contains three convolutional blocks, each comprising a 2D convolution layer with increasing filter sizes of 16, 32, and 64, respectively.

Each convolutional layer is followed by batch normalization and a ReLU activation to stabilize and introduce non-linearity to the learning process. Pooling layers are used after each convolutional block to reduce the spatial dimensions, with max pooling applied in the first two blocks and average pooling in the final block. After feature extraction, the features are passed to a fully connected layer of 128 units, producing a compact feature vector that encodes the essential spatial characteristics of the input spectrograms.

B. LSTM Architecture

The LSTM network is employed as a temporal classifier to process the feature vectors extracted by the CNN. A sequence input layer of size 128 receives the output from the CNN's fully connected layer, treating each feature vector as a time step in a sequence. The LSTM layer, configured with 256 hidden units, is capable of learning the temporal dependencies and sequential patterns in the feature space. The LSTM's output is passed through a fully connected layer corresponding to the number of output classes, followed by a softmax activation for multi-class probability distribution and a classification layer for final label prediction.

C. Model Overview

The overall model combines the strengths of CNNs and LSTMs, where the CNN efficiently extracts spatial features from spectrogram images representing EEG data, and the LSTM models the temporal relationships between the extracted features. This hybrid CNN-LSTM architecture is specifically designed to classify motor imagery tasks from EEG data, leveraging both the spatial patterns in the spectrogram and the temporal evolution of these patterns to make accurate predictions.

D. Training and Testing

The data is split into 70% Training, 20% Testing and 10% for validation. The CNN and LSTM networks are trained in two stages. First, the CNN is trained for 30 epochs using the Adam optimizer with a learning rate of $1e-4$, and validation is performed at regular intervals to monitor overfitting. Once the CNN is trained, the extracted features are extracted by unfreezing the final layer, reshaped into sequences and fed into the LSTM, which is trained for 70 epochs with similar optimization settings. The model's performance is evaluated on a test dataset using classification accuracy and confusion matrices to assess the quality of predictions across different motor imagery tasks.

V. ROBOTIC ARM INTEGRATION

A. 4-DOF Configuration

The 4-DOF robotic arm enables left-right, up-down, forward-backward, and grip motions, mimicking key upper-limb functions for rehabilitation tasks. Controlled via motor imagery decoded from EEG using a CNN-LSTM model, it operates in real time through an Arduino Uno and SG90 servo motors. Each DOF is mapped to mental commands

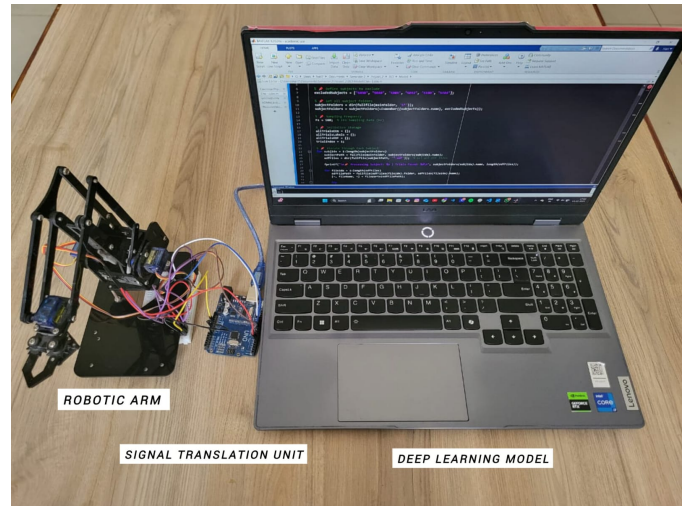


Fig. 6. Experimental Setup

(e.g., "Left", "Right", "Rest"), allowing hands-free neuro-rehabilitation. Fig. 6 illustrates the setup.

B. Signal Translation Unit

SG90 servo motors were chosen for their compact size, high torque (11 kg-cm at 6V), and precision, ideal for simulating joint movements in rehabilitation. Controlled by an Arduino Uno serves as the control hub, receiving motor imagery labels ("LEFT", "RIGHT", "REST") from MATLAB via serial communication. It maps these labels to predefined PWM signals, driving SG90 servo motors for precise joint movements. The PWM signal wires are connected to digital I/O pins D3, D5, D6, and D9, which are used to send control pulses that dictate servo angles. This configuration enables independent and precise control of each joint in the 4-DOF robotic arm, ensuring synchronized and accurate actuation based on the motor imagery classification output. Fig. 7 represents the input from the model to arduino.

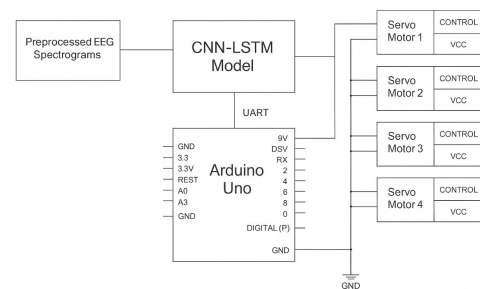


Fig. 7. Input from the model to arduino

VI. RESULT

The combination of MATLAB for motor imagery-based signal processing and classification with Arduino to actuate a robotic arm provides a smooth pipeline from idea to movement. The arm, with SG90 servo motors having high torque,

answers motor imagery cues precisely, providing natural arm imitation and a real-world meaningful treatment tool.

The performance of the proposed CNN-LSTM model is assessed using evaluation metrics including accuracy, confusion matrix, and class-wise precision, recall, and F1-score. The results indicate robust classification performance across the three motor imagery classes—Left, Right, and Rest—demonstrating the model’s effectiveness in decoding motor intentions from EEG signals. The training progression and convergence behavior of the model are depicted in Fig. 8, highlighting stable learning dynamics and effective optimization.

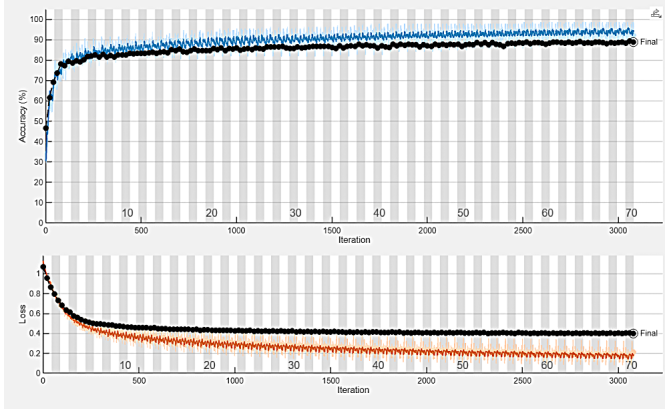


Fig. 8. Training Accuracy/Loss vs iteration

A. Model Analysis

The proposed hybrid CNN-LSTM model demonstrated effective classification performance on the motor imagery EEG spectrogram dataset. The CNN model served as a feature extractor, achieving a validation accuracy of **83.69%**. When combined with an LSTM network for temporal modeling, the overall classification performance improved to **85.89%**. The training and validation accuracy and loss curves depicts that the models learned meaningful features from the data.

TABLE I
PRECISION, RECALL, AND F1-SCORE FOR EACH CLASS

Class	Precision	Recall	F1-Score
T0	0.88	0.88	0.88
T1	0.75	0.76	0.75
T2	0.82	0.81	0.82

B. Confusion Matrix Analysis

- The **T0 class** (Rest) was correctly classified 116 times. It was occasionally misclassified as T1 (5 instances) and T2 (8 instances).
- The **T1 class** (Left) had 39 correct predictions, with confusion occurring as T0 (6 instances) and T2 (4 instances).
- The **T2 class** (Right) was accurately identified 52 times, with minor misclassifications as T0 (5 instances) and T1 (6 instances).

Overall, as illustrated in Fig. 9 the model demonstrates good discriminative capability across all classes, with the majority of samples being classified correctly. Minor confusion between neighboring motor imagery classes is expected in EEG-based classification tasks due to overlapping neural activations.

True Class	T0	116	5	8
	T1	6	39	4
	T2	5	6	52
	Predicted Class	T0	T1	T2

Fig. 9. Confusion Matrix

VII. ABLATION STUDY

To evaluate the contribution of individual components within the architecture, an ablation study was performed by systematically modifying or removing key elements. The following variants were tested:

- **CNN only:** Classification using the CNN backbone without the LSTM component.
- **Shallow CNN:** Reduced CNN with fewer convolutional layers to assess depth sensitivity.
- **Shallow LSTM:** LSTM with fewer units trained on CNN-extracted features to test temporal modeling depth.
- **No Batch Normalization:** CNN-LSTM trained without batch normalization layers to examine regularization impact.

These configurations were trained and evaluated using the same data splits, optimizer, and hyperparameters to ensure fair comparison. The results are summarized in Table II.

TABLE II
ABLATION STUDY RESULTS FOR CNN-LSTM MODEL

Model Variant	Accuracy (%)
CNN-LSTM	85.89
CNN Only	82.20
Shallow CNN	70.55
Shallow LSTM	82.10
No Batch Normalization	77.54

VIII. DISCUSSION

The proposed model exhibits low computational latency, making it suitable for real-time applications. The CNN feature extractor took approximately 4 minutes and 56 seconds to train, while the LSTM classifier completed training in just 35 seconds, highlighting the model’s efficiency. This rapid training is attributed to the use of only three EEG channels

(C3, Cz, and C4), along with a compact yet effective CNN-LSTM architecture. The reduced model complexity not only accelerates training and inference but also facilitates deployment on low-power hardware.

The overall system demonstrates a fully data-driven BCI-based robotic arm framework for motor rehabilitation. Previous works have explored low-cost robotic arm control using EEG through switch- and thought-based mechanisms [11], as well as EEG-driven prosthetic arm designs emphasizing motor task pattern extraction and neuro-inspired processing [10].

Cong Li et al. [3] proposed a hybrid neural network (HNN) combining CNN, LSTM, and DNN for classifying motor imagery EEG signals using data from 12 subjects and 64 electrodes (excluding 11 based on the 10–10 system) from the PhysioNet dataset. The model targets five motor tasks and integrates spatial, temporal, and high-level features. Achieving 75.52% accuracy, it outperformed traditional methods such as Common Spatial Pattern with Linear Discriminant Analysis (CSP+LDA) and Filter Bank Common Spatial Pattern (FBCSP), demonstrating enhanced generalization and robustness in EEG-based BCI applications.

The paper [2] introduced a novel method of transforming EEG signals into RGB-like 2D images by encoding Delta, Mu, and Beta bands. A DCNN-LSTM model was applied to these spectrograms, achieving a classification accuracy of 70.64%, outperforming the traditional SVM baseline by 5%. This highlights the effectiveness of combining frequency-domain encoding with deep learning for multi-class MI classification.

Alnaanah et al. [6] compared CNN1D, CNN2D, CNN3D, and CNN1D_MF architectures for motor imagery classification on the PhysioNet dataset (64 electrodes, 5 classes). The CNN1D_MF model, using multiple frequency band filters (0.5–40 Hz), achieved the best accuracy (58.0%), highlighting the advantage of frequency-domain augmentation for cross-subject generalization.

TABLE III
COMPARISON OF MOTOR IMAGERY EEG CLASSIFICATION MODELS

Model	Acc. (%)	Subjects, Electrodes, and Metrics
CNN-LSTM (This Work)	85.89	30 subjects, 3 electrodes
CNN-LSTM	70.64	109 subjects, 64 electrodes [2]
CNN+LSTM+DNN	75.52	12 subjects, 64 electrodes [3]
CNN1D_MF	58.0	105 subjects, 64 electrodes [6]

The comparative analysis in Table III highlights that the proposed CNN-LSTM model, despite utilizing only three electrodes (C3, Cz, C4) and a smaller subject pool, surpasses the performance of more complex models such as DCNN-LSTM and hybrid CNN-LSTM-DNN frameworks that use higher electrode counts and larger datasets. This demonstrates the model's efficiency and robustness in decoding motor imagery tasks while significantly reducing computational and hardware complexity, making it ideal for real-time and low-resource BCI applications.

IX. CONCLUSION AND FUTURE SCOPE

This work presents a non-invasive Brain-Computer Interface framework for motor rehabilitation using EEG-based motor imagery classification. EEG signals from 30 subjects were acquired from three key electrodes (C3, Cz, C4), preprocessed to remove artifacts and converted into 2D spectrograms via Short-Time Fourier Transform. A hybrid Convolution neural network-Long short term memory was employed CNN for spatial feature extraction and LSTM for modeling temporal dynamics—to classify imagined movements. The classified outputs were transmitted to an Arduino Uno to control a 4-degree-of-freedom robotic arm, enabling real-time execution of intended motor actions. This closed-loop system offers a low-cost, hardware-efficient, and adaptive rehabilitation solution.

The proposed system can be further extended by incorporating a larger subject pool to enhance model robustness and generalization. Additionally, integration with real-time EEG acquisition using consumer-grade or clinical headsets will enable practical validation and deployment in real-world rehabilitation environments.

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